

# AMIN RANEM, PhD

Machine Learning Researcher · Medical Imaging and Clinical AI

Enschede, the Netherlands · [a.ranem@utwente.nl](mailto:a.ranem@utwente.nl) · +49 1525 5383430 · [Website](#) · [Google Scholar](#) · [LinkedIn](#) · [GitHub](#)

Machine learning researcher with a PhD in Computer Science, specializing in medical imaging AI. I work on continual learning, multi-modal models that combine clinical data with imaging, and adapting foundation models to medicine. My 13 peer-reviewed publications appeared at CVPR, WACV, MICCAI, MIDL, BMVC and in Nature Scientific Reports. I also care about what happens after publication. I ran quality assurance across 38 German university hospitals ([RACOON](#)), studied FDA regulatory research (PCCP), and currently prepare models for prospective clinical validation with surgeons and industry partners. Recent A&I organizer in the MICCAI community and invited speaker at Stanford.

## CORE EXPERTISE

**Machine Learning** Continual and lifelong learning · multi-modal learning (imaging + structured clinical data) · medical image segmentation and registration · foundation models (SAM) · Neural Cellular Automata · transformers · explainable AI (XAI)

**Engineering** Python · PyTorch · Git · Linux · reproducible training and evaluation pipelines · large-scale imaging data (CT, MRI, CTA)

**Clinical & Regulatory** Prospective multi-center validation studies · quality assurance for clinical AI at national scale · federated learning environments · FDA PCCP and software as a medical device

## EXPERIENCE

**Postdoctoral Researcher, Medical AI** · University of Twente, Mathematics of Imaging & AI, NL Nov 2025 – Present

- Design, implement and train **two multi-modal AI risk models** for the AI4EVAR project. The models combine structured clinical data with CTA imaging to predict re-interventions after endovascular aneurysm repair (ZonMw-funded consortium of three hospitals, two universities and industry partners).
- Train on the RADAR database of **1,000+ EVAR patients**. Both models will be validated prospectively with around 300 patients across three hospitals.
- Embed **explainable AI** into both models so vascular surgeons can understand each multi-modal prediction. The outputs feed shared decision tools developed with clinicians and **Pie Medical Imaging**.

**Doctoral Researcher / Research Assistant** · TU Darmstadt, MEC-Lab, DE Apr 2022 – Jun 2025

- Published **13 peer-reviewed papers** on continual learning for medical image analysis (CVPR, WACV, MICCAI, MIDL, BMVC, Nature Scientific Reports). PhD completed in 3 years.
- Held a key quality assurance role in **RACOON**, Germany's national radiology AI platform. Built QA methods for CT/MRI images and segmentation masks used across **38 university hospitals**.
- Researched regulatory pathways for continually learning medical software and co-authored a peer-reviewed analysis of the **FDA PCCP** framework.
- Supervised BSc and MSc theses and DAAD-WISE research interns. Taught the VAE lecture in Deep Generative Models and the Registration and Continual Learning lectures in Deep Learning for Medical Imaging every year.

**Earlier Roles** · TU Darmstadt · Deutsche Lufthansa AG · DFKI 2018 – 2022

- Student assistant for RACOON quality assurance at TU Darmstadt (2020–2022).
- Business intelligence intern at Deutsche Lufthansa AG, building Qlik Sense analytics (2019–2020).
- Student assistant for prototype development at the German Research Center for Artificial Intelligence, DFKI (2018–2019).

## EDUCATION

**Ph.D., Computer Science** · TU Darmstadt, DE 2022 – 2025

- Thesis: **"Beyond End-to-End — Exploring Extremes in Medical Continual Learning."**

**M.Sc., Business Informatics** · TU Darmstadt, DE 2020 – 2022

- Thesis: "Continual Learning with Transformer Architectures."

**B.Sc., Business Informatics** · Saarland University, DE 2017 – 2020

## SELECTED PROJECTS

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- **RACoon quality assurance.** Built image and segmentation QA pipeline behind Germany's national radiology AI platform, keeping data from 38 hospitals usable for federated research.
- **Lifelong nnU-Net.** Co-developed an open framework that makes continual learning experiments on medical segmentation reproducible and comparable across methods, using the SOTA nnU-Net as backbone. Published in Nature Scientific Reports.
- **Neural Cellular Automata for medical imaging.** Showed with NCAdapt and NCA-Morph that models a fraction of the size of a U-Net can segment and register medical images. Follow-up work trained them directly on edge devices (MICCAI 2024).
- **Foundation models in medicine.** Adapted the Segment Anything Model to continual prostate MRI segmentation (UnCLE SAM, MIDL 2024) and studied its behavior on radiology and pathology data.

## PUBLICATIONS

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Also listed on [Google Scholar](#).

- **A. Ranem**, J. Kalkhof, A. Mukhopadhyay. "NCAdapt: Dynamic adaptation with domain-specific Neural Cellular Automata for continual hippocampus segmentation." **WACV**, 2025.
- S.K. Sivakumar, Y. Frisch, **A. Ranem**, A. Mukhopadhyay. "SASVi: Segment any surgical video." **Int. Journal of Computer Assisted Radiology and Surgery**, 2025.
- N. Babendererde, **A. Ranem**, M. Fuchs, C. Gonzalez, H.J. Krumb, A. Mukhopadhyay. "FDA's PCCP: Opportunities and Gaps." **Workshop on Bridging Regulatory Science & Medical Imaging Evaluation**, 2025.
- **A. Ranem**, J. Kalkhof, A. Mukhopadhyay. "NCA-Morph: Medical Image Registration with Neural Cellular Automata." **BMVC (Oral)**, 2024.
- J. Kalkhof, **A. Ranem**, A. Mukhopadhyay. "Unsupervised Training of Neural Cellular Automata on Edge Devices." **MICCAI**, 2024.
- **A. Ranem**, M.A.M. Aflal, M. Fuchs, A. Mukhopadhyay. "UnCLE SAM: Unleashing SAM's Potential for Continual Prostate MRI Segmentation." **MIDL**, 2024.
- **A. Ranem**, C. González, D. Pinto dos Santos, A.M. Bucher, A.E. Othman, A. Mukhopadhyay. "Continual atlas-based segmentation of prostate MRI." **WACV**, 2024.
- **A. Ranem**, N. Babendererde, M. Fuchs, A. Mukhopadhyay. "Exploring SAM Ablations for Enhancing Medical Segmentation in Radiology and Pathology." **arXiv preprint**, 2023.
- C. González, **A. Ranem**, D. Pinto dos Santos, A. Othman, A. Mukhopadhyay. "Lifelong nnU-Net: a framework for standardized medical continual learning." **Nature Scientific Reports**, 2023.
- **A. Ranem**, C. González, A. Mukhopadhyay. "Continual Hippocampus Segmentation with Transformers." **CVPR Workshops**, 2022.
- C. González, **A. Ranem**, A. Othman, A. Mukhopadhyay. "Task-Agnostic Continual Hippocampus Segmentation for Smooth Population Shifts." **DART Workshop, MICCAI**, 2022.
- **A. Ranem**, J. Kalkhof, C. Özer, A. Mukhopadhyay, I. Oksuz. "Detecting respiratory motion artefacts for cardiovascular MRIs to ensure high-quality segmentation." **STACOM Workshop**, 2022.
- C. González, C.L. Harder, **A. Ranem**, R. Fischbach, I.J. Kaltenborn, A. Dadras, A.M. Bucher, A. Mukhopadhyay. "Quality monitoring of federated Covid-19 lesion segmentation." **Bildverarbeitung für die Medizin**, 2022.

## INVITED TALKS

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- **Stanford CNS Lab:** "Exploring the extremes of medical continual learning" (*Sep 2024*)
- **BMVC 2024, Oral Session:** "NCA-Morph: Medical Image Registration with Neural Cellular Automata" (*Nov 2024*)
- **MICCAI Tutorial, Data-Centric Dynamic Learning:** "Hands-on: Continual Learning with UnCLE SAM" (*Oct 2024*)
- **CapsNetwork School:** "Beyond end-to-end continual segmentation" (*Nov 2023*)

## LEADERSHIP & COMMUNITY

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- **MICCAI Student Board, Professional Events Officer (2024–2025).** Organized the Academia & Industry event, which brings researchers and industry into the same room.
- **Workshop and tutorial organizer.** DGM4MICCAI workshop (MICCAI 2023–present) · Data-Centric Dynamic Learning tutorial (MICCAI 2024) · CapsNetwork school, Darmstadt (2023).
- **Mentoring.** Guided BSc and MSc thesis students and international DAAD-WISE interns at TU Darmstadt and University of Twente (2022–present).

## LANGUAGES

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English (native) · German (native) · Arabic (intermediate) · French (basic)